ENCORE 0.41 · HOW IT WORKS

How Encore recommends music

Smart shuffle, taste scores, likes, skips and dislikes, the “For you” mixes, Start radio and Genre shuffle, explained step by step with diagrams and worked numbers.

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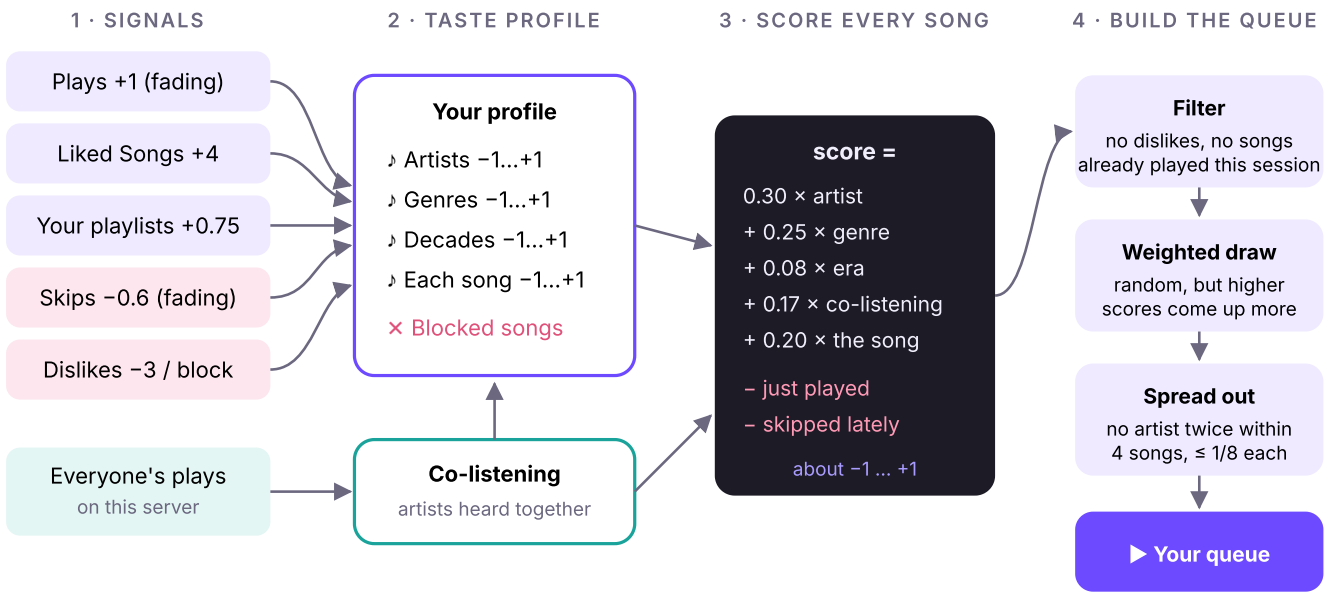
Everything stays on your server

Encore works out recommendations from your own library and your own listening only. No outside service is asked anything, and every number in this guide is taken from the code that ships in Encore (the server's taste engine, `server/crates/library/src/taste.rs`). The worked example was checked by running that code.

Written 26 October 2026 for Encore 0.41.1.

1 The big picture

Every recommendation in Encore, from pressing Shuffle to a “For you” mix, goes through the same four steps.



The profile is made fresh on every request, from what is in the database at that moment.

In plain words

1. **Signals.** Encore notices what you play, like, put on playlists, skip and dislike.
2. **Profile.** Each signal is credited not just to the song but also to its artists, its genres and its decade. Added up, that gives a number between −1 (you avoid it) and +1 (you love it) for every artist, genre and decade you've touched. Encore also knows which artists people on your server tend to play together.
3. **Score.** Every song in the library gets a score from those numbers. Songs you heard in the last few hours, or skipped lately, lose points for a while.
4. **Queue.** Songs are picked at random, but a high score makes a song more likely to be picked. Songs you've disliked are never picked. Finally the list is reordered so the same artist doesn't come up again within the next three songs.

Why random at all?

If Encore always played the highest scores it would play the same 50 songs forever. A *weighted* draw means favourites come up more often, but any song you haven't ruled out can still turn up. That's how you rediscover things.

Brand new listener?

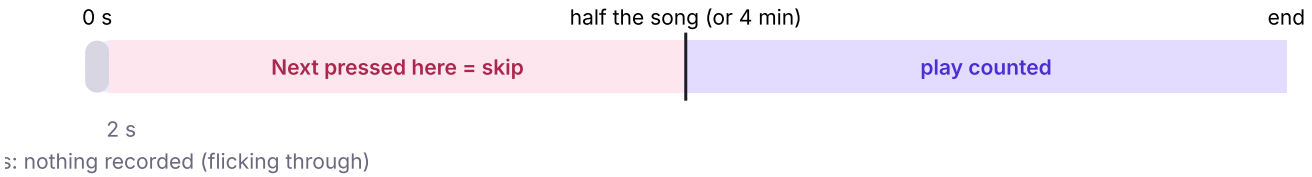
With no plays, likes, skips or dislikes yet, every song scores the same and smart shuffle is simply a plain shuffle (still with artists spread out). It starts adapting from the first counted play.

2 What Encore pays attention to

Six kinds of signal. Each has a weight (how much it says about your taste) and a lifetime.

SIGNAL	WEIGHT	LIFETIME	DETAILS
Play	+1.0	Halves every 120 days	Counts once you've heard half the song or 4 minutes, whichever comes first (very short songs: to the end).
Song in Liked Songs	+4.0	Never fades	The strongest “yes”. Liking a song also takes back a dislike of it.
Song on one of your own playlists	+0.75	Never fades	Each song once, however many playlists it is on. Playlists shared with you don't count.
Skip	−0.6	Halves every 21 days, ignored after 90	Pressing Next after 2 seconds but before the play counted. A mild “not now”.
Dislike (a song)	−3.0	Until you take it back	The song is never played by any recommendation again, and leaves Liked Songs.
Don't play this artist	blocks	Until you take it back	Every song where they are the main artist is never played. Songs they only feature on just sink.

When is it a play, and when a skip?



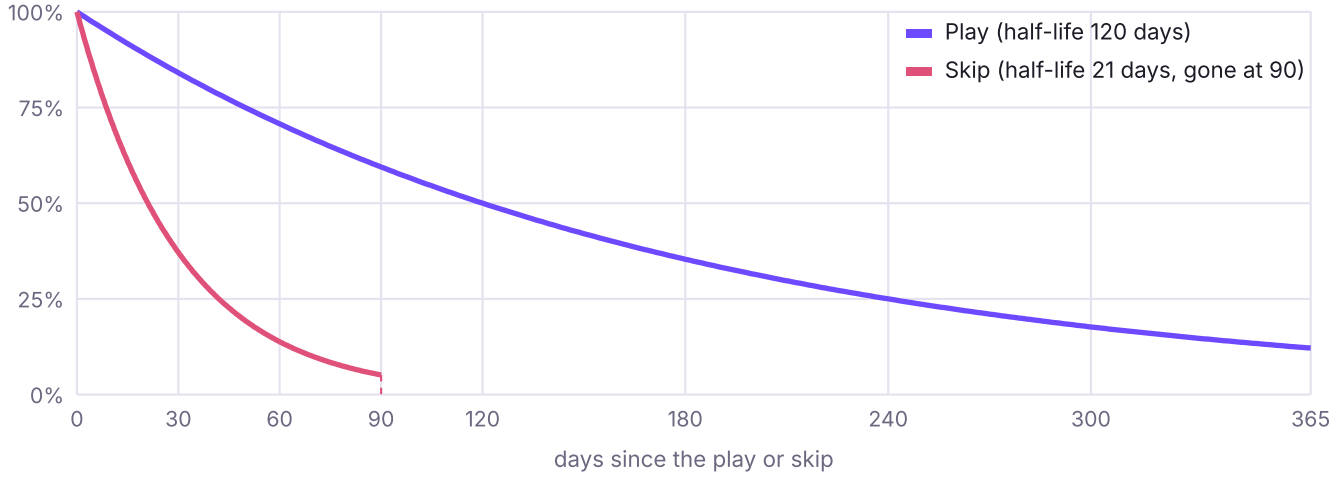
- A song counts as **played** once you've heard half of it or 4 minutes, whichever comes first (the same rule Last.fm uses). Songs under 30 seconds count at the end.
- Pressing **Next** after the first 2 seconds but before the play counted is a **skip**. Songs ending on their own are never skips, and Next on a disliked song isn't sent as a skip (the dislike already says more).
- Encore reads your plays of the last 400 days (up to 20,000 of the newest) and your skips of the last 90 days. Older skips are deleted.

Dislikes and likes cancel each other

Disliking a song removes it from Liked Songs; liking a disliked song takes the dislike back. Both kinds of dislike can be undone at any time, and the song or artist immediately becomes playable again.

3 Memory fades

Taste drifts. A play from last week says more about you than one from last year, so every play and skip counts less as it gets older.



How much one play or skip still counts, by age. Plays halve every 120 days. Skips are forgotten much faster: they halve every 21 days and stop counting at all after 90.

AGE	A PLAY COUNTS	A SKIP COUNTS
0 days	100%	100%
7 days	96%	79%
21 days	89%	50%
30 days	84%	37%
60 days	71%	14%
90 days	59%	5%
120 days	50%	ignored
240 days	25%	ignored
365 days	12%	ignored

Example: a play

You played a song 30 days ago. It still adds **+0.84** instead of +1.0. A year-old play adds only **+0.12**.

Example: a skip

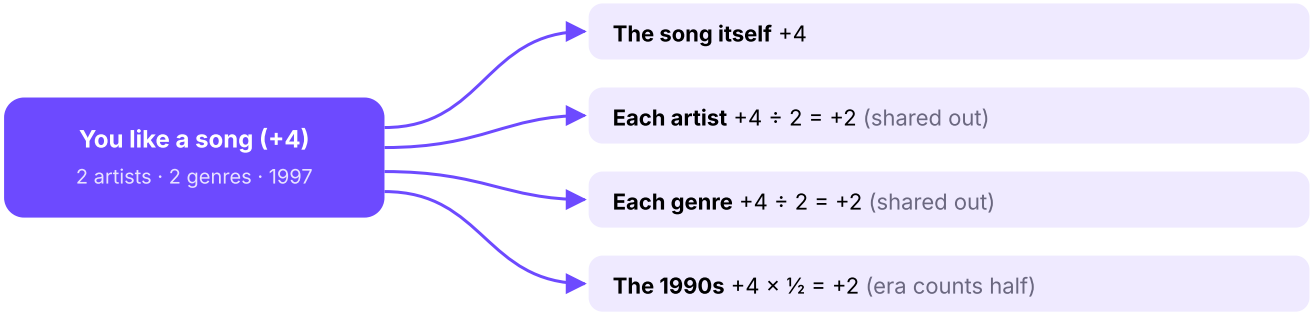
You skipped a song 7 days ago. It takes off 0.6 × 79% = **−0.48**. Three weeks later the same skip is only worth −0.15, and after 90 days it's gone.

Likes never fade

A song in Liked Songs is always worth +4, however long ago you liked it.

4 Your taste profile

Each signal is credited to four places at once: the song, its artists, its genres and its decade.



Sharing out means a song tagged with five genres doesn't count five times as much as a song with one. Songs with no genre tag borrow the two genres their main artist's other songs use most, so they still land in the right genre mixes.

Squeezing totals into −1 … +1

After adding everything up, one favourite artist might have a total of 200 while others have 3. To stop one giant from drowning everyone else out, totals are squeezed onto a −1 … +1 scale:

```
affinity = tanh( 1.5 × total ÷ typical )
typical = the average of your ten biggest positive totals (at least 1)
tanh    = a smooth curve that flattens out near -1 and +1
```

A song's own total is squeezed with a fixed scale instead: `tanh(total ÷ 3)`. So one like (+4) already makes a song's own value +0.87.

Worked example: meet Sam

Sam's history, which the rest of this guide follows:

- **A** Radiohead “Paranoid Android” (1997, alternative rock): played every 3 days for two months (19 plays), liked, on a playlist.
- **B** Radiohead “Idioteque” (2000): on a playlist, never played.
- **D** Portishead “Glory Box” (1994, trip hop): liked, on a playlist, 5 plays about six months ago, and played again **2 hours ago**.
- **E** Taylor Swift “Blank Space” (2014, pop): skipped 2 and 5 days ago.
- **H** Muse “Starlight” (2006, alternative rock): disliked.
- **C** Massive Attack “Teardrop”, **F** Miles Davis “So What” and **G** Muse “Hysteria”: never touched.

ARTIST	TOTAL	AFFINITY
Radiohead	+21.55	+0.98
Portishead	+7.63	+0.66
Taylor Swift	-1.07	-0.11
Muse	-3.00	-0.30

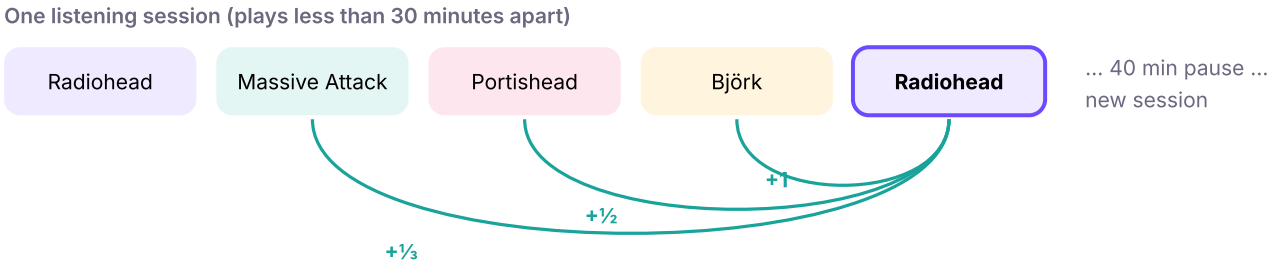
GENRE	TOTAL	AFFINITY
alternative rock	+18.55	+0.97
trip hop	+7.63	+0.70
pop	-1.07	-0.12

DECADE	TOTAL	AFFINITY
1990s	+14.21	+0.91
2000s	-1.12	-0.12
2010s	-0.54	-0.06

Radiohead's total is 21.55: 19 fading plays (≈ 16.8) + like 4 + two playlist songs 1.5. The “typical” value here is 14.59, so tanh(1.5 × 21.55 ÷ 14.59) = 0.98. Muse is negative because of the one disliked song, yet alternative rock as a genre stays strongly liked.

5 Co-listening: artists people play together

Encore learns which artists belong together from how people on your server actually listen. With one listener it learns from your own sessions; with several it becomes a small “people who play X also play Y”.



Each play is linked to the three songs before it in the same session: the one right before counts 1, two back ½, three back ⅓. The counts for every pair of artists, across everyone's plays of the last year, become a similarity between 0 and 1:

```
similarity(X, Y) = together ÷ √(plays of X × plays of Y)  ×  together ÷ (together + 1)
the second part damps pairs heard together only once or twice
```

Example

Radiohead was played 20 times, Massive Attack 10 times, and the “together” count is 6. Similarity = 6 ÷ √200 × 6 ÷ 7 = 0.42 × 0.86 = **0.36**. Each artist keeps its 40 closest neighbours. This is rebuilt at most every 10 minutes.

How it reaches your profile

For every artist you like, Encore adds *your liking × their similarity* to each neighbour, then divides by the biggest result so the top neighbour is 1. That's the **co-listening** value in the score: it lets an artist you've never played come up because it sits next to artists you love.

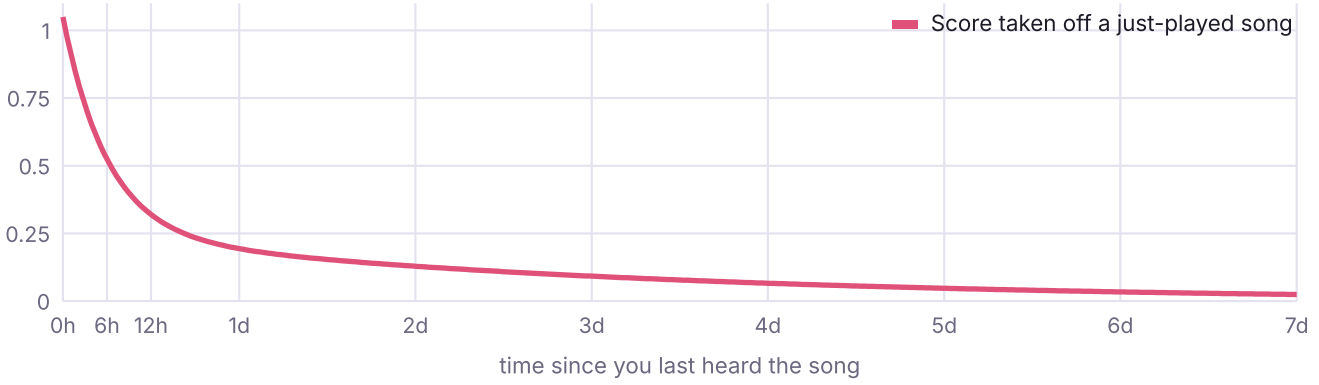
In Sam's example we suppose the server has learned: Radiohead–Massive Attack 0.40, Portishead–Massive Attack 0.55, Radiohead–Muse 0.30, Radiohead–Portishead 0.20. Massive Attack gets 0.98 × 0.40 + 0.66 × 0.55 = 0.75, the highest, so its co-listening value is **1.00**; Muse gets **0.39**.

6 Scoring a song

Every song gets one number, roughly between −1 and +1. Higher means a better fit for you right now.

```
score = 0.30 × artist         average affinity of the song's artists
      + 0.25 × genre          average affinity of its genres
      + 0.08 × era            affinity of its decade
      + 0.17 × co-listening   best co-listening value among its artists
      + 0.20 × the song       its own history (plays, like, playlists, skips)
      - just-played penalty
      - 0.25 × recent skips   the faded skip count
```

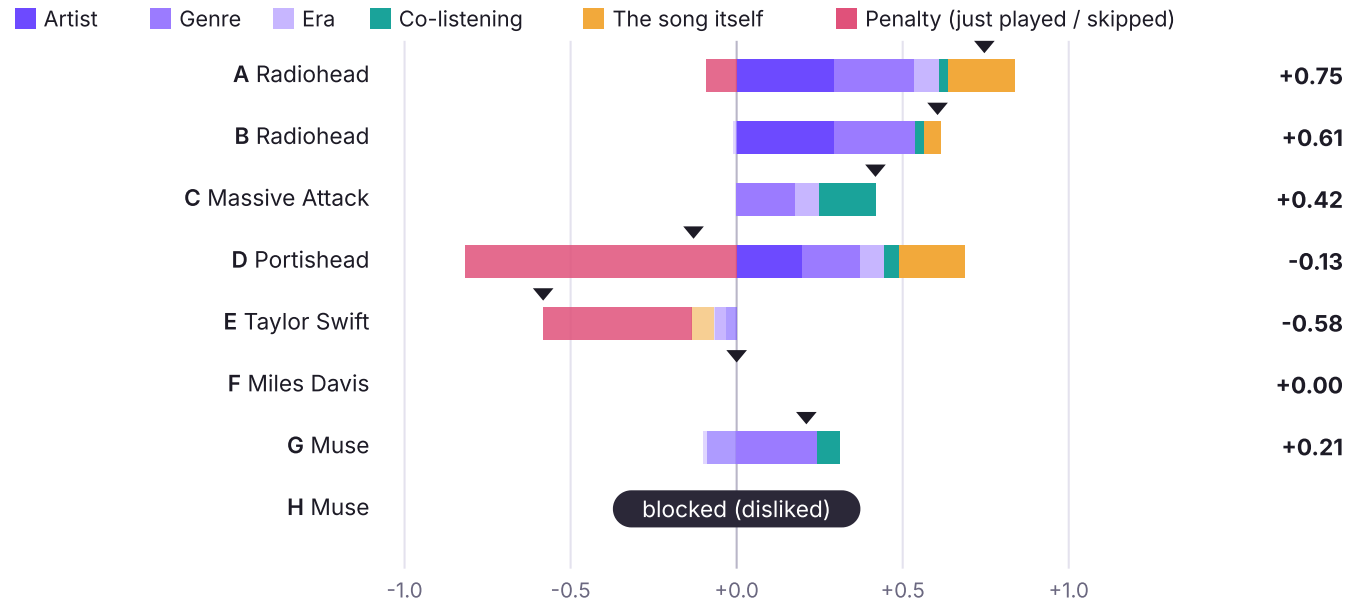
The just-played cooldown



just now	1 hour	3 hours	6 hours	12 hours	1 day	2 days	3 days	1 week
−1.05	**−0.92**	**−0.73**	**−0.52**	**−0.32**	**−0.19**	**−0.13**	**−0.09**	**−0.02**

A strong dip for the next few hours (0.8, fading over ~6 hours) plus a gentle one for the next few days (0.25, fading over ~3 days). On top of that, the player never repeats a song within the same listening session until everything else has played.

Sam's scores



SONG	ARTIST	GENRE	ERA	CO-L.	SONG	PENALTY	SCORE	1ST PICK
A Radiohead “Paranoid Android”	+0.293	+0.243	+0.072	+0.030	+0.200	−0.092	**+0.75**	43.9%
B Radiohead “Idioteque”	+0.293	+0.243	-0.009	+0.030	+0.049	0	**+0.61**	27.5%
C Massive Attack “Teardrop”	0	+0.176	+0.072	+0.170	0	0	**+0.42**	14.7%
D Portishead “Glory Box”	+0.197	+0.176	+0.072	+0.044	+0.198	−0.816	**-0.13**	2.4%
E Taylor Swift “Blank Space”	-0.033	-0.031	-0.005	0	-0.068	−0.446	**-0.58**	0.5%
F Miles Davis “So What”	0	0	0	0	0	0	**+0.00**	3.7%
G Muse “Hysteria”	-0.090	+0.243	-0.009	+0.066	0	0	**+0.21**	7.4%
H *Muse “Starlight”*	*Disliked: never played, not even scored*							

- **A** is the favourite (+0.75) even though it was heard 3 days ago (−0.09).
- **B** was never played, but Radiohead and alternative rock carry it to second place. This is how you get more of what you like.
- **C** Massive Attack has never been played by Sam, but trip hop, the 1990s and co-listening lift it well above neutral: a new find.
- **D** is a liked favourite, but it played 2 hours ago, so it loses 0.82 and drops below zero. In a day or two it's back near the top.
- **E** Two skips this week (faded weight 1.78) cost 0.45 on top of the small negatives from the skips themselves.
- **F** Miles Davis has nothing for or against it: exactly 0. Still playable, just less often.
- **G** Muse is held back by the disliked Muse song, but the alternative rock genre keeps it positive.

7 The never-play rules

Scores decide how *often*. These rules decide *never*: a blocked song isn't even scored, so no amount of luck brings it back.

RULE	WHAT IS BLOCKED	HOW TO UNDO
Disliked song	That song, in every shuffle, mix, radio and in “keep going”. Disliking the song that's playing moves straight on to the next one (not counted as a skip). It also pulls its artist, genres and decade down by 3 (shared out), so similar songs come up less.	Take the dislike back, or like the song.
Don't play this artist	Every song where the artist is the *main* artist. Songs where they're only featured stay playable but rank far lower.	Take it back from the dislikes list.
Skipped again and again	A song whose faded skips add up to 2.5 or more is left out until the skips fade below that.	Happens on its own, or like the song: a liked song is never dropped for skips, it only sinks for a while.
Already played this session	“Keep going” and Shuffle never repeat a song you've heard (or that's already queued) in this listening session, until the whole library has played.	Starts over automatically when everything has played.

Example: when do skips block a song?

Three skips today: 1 + 1 + 1 = 3.0 → **blocked**.
Three skips a week ago: 3 × 0.79 = 2.38 → still allowed, but its score is down by 0.25 × 2.38 = 0.60.
Four skips a week ago: 4 × 0.79 = 3.17 → **blocked** for about another week, until they fade below 2.5.
Sam's two skips of “Blank Space” add up to 1.78: not blocked, just pushed down.

Plain shuffle respects dislikes too

Even with smart shuffle switched off (Settings → Playback), the plain library shuffle still leaves out disliked songs and artists. Only the taste weighting is switched off.

8 Picking the queue

Step 1: the weighted draw

Each allowed song gets a weight from its score. Then songs are drawn one after another, without repeats, each time with a chance proportional to its weight.

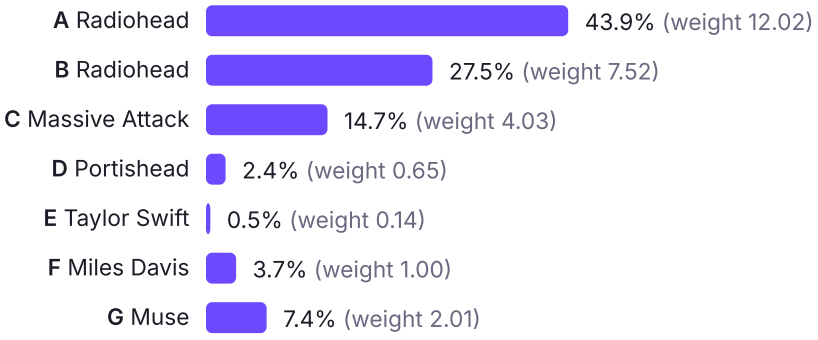
```
weight = e^(score ÷ 0.3)      0.3 is the "temperature": lower follows the scores more strictly,
                                 higher is closer to a plain shuffle
```

SCORE	WEIGHT (VS. A NEUTRAL SONG)
-0.5	0.19×
-0.2	0.51×
+0.0	1.00×
+0.2	1.95×
+0.4	3.79×
+0.6	7.39×
+0.9	20.09×

A song scoring +0.9 is about **20 times** as likely to be picked as a song you've never touched (score 0), and a song at −0.5 about **5 times less** likely. Nothing allowed ever has zero chance.

Technically this is the Efraimidis–Spirakis method (equivalently "Gumbel top-k"): each song gets a random key from its weight and the best keys win, which is the same as drawing one by one.

Sam's first pick



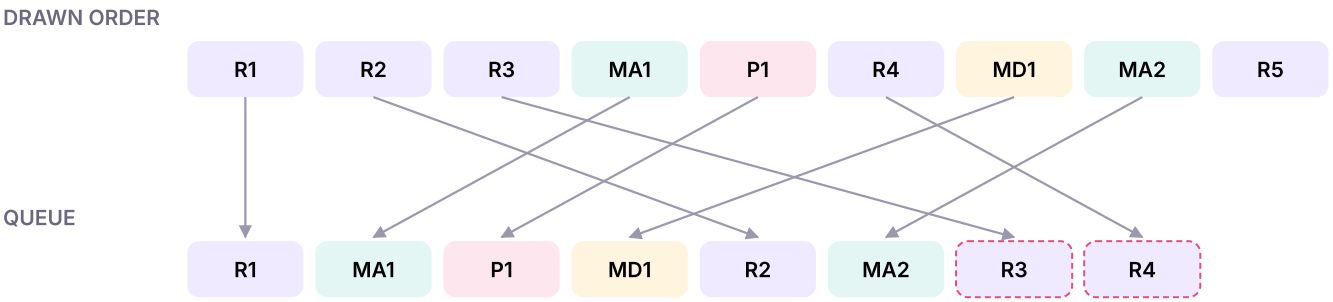
The chance each song is picked first out of Sam's seven allowed songs. A real library has thousands of songs, so each one's share is much smaller, but the ratios are the same: "Paranoid Android" is 12× as likely as Miles Davis, and Miles Davis is still 7× as likely as the twice-skipped Taylor Swift song.

Step 2: spreading artists out

The draw gives an order with favourites near the front, which could put three Radiohead songs in a row. Encore then walks down that order and, for each place, takes the first song that breaks none of these rules:

1. Its main artist isn't one of the last three songs' main artists (so an artist comes back at most once every four songs).
2. No artist, and no album, takes more than an eighth of the list (at least 2 songs each; 6 in a 50-song list).

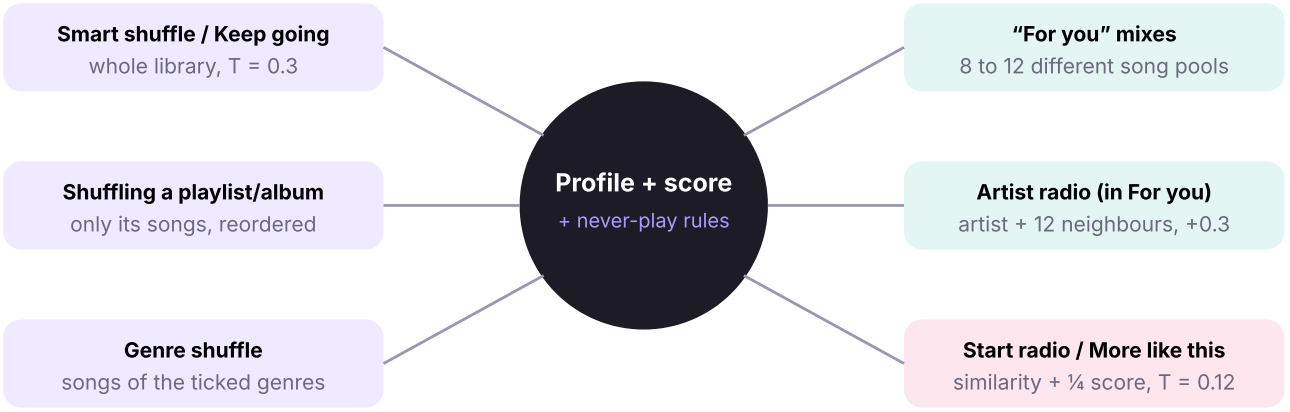
If nothing in the next 400 songs fits, rule 2 is relaxed first, then rule 1, so a list made of only one artist still fills up.



A tiny 8-song example (R = Radiohead, MA = Massive Attack, P = Portishead, MD = Miles Davis; R1, R2 and R4 are on the same album). Radiohead is spaced out every four songs. At the end only Radiohead is left and Radiohead already has its 2-song share, so both rules are eased and R3, R4 (dashed) fill the last places. With a real library there are almost always other artists to pick, so this rarely happens.

9 Every feature, and how it uses the same engine

All features share one profile and one scoring formula. They differ only in **which songs** are in the running and **how strictly** the score is followed.



Smart shuffle and “keep going”

On by default (Settings → Playback → Smart shuffle). The Shuffle button asks for 50 songs from the whole library, drawn as in section 8. When your queue runs out, Encore keeps the music going with another 50 the same way, leaving out every song already played or queued this session. If you switch smart shuffle off, both become a plain random shuffle (dislikes still left out).

Shuffling a playlist or album

Turning shuffle on for a list first shuffles it on your device instantly, then asks the server for a taste-weighted order of the same songs: favourites tend to come earlier, the same artist is spread out, disliked songs drop out. If the server can't answer, or the queue changed meanwhile, the plain shuffle stays.

Genre shuffle

Pick one or more genres (your favourite genres are listed first, each with its song count; genres with fewer than 3 songs aren't listed). Encore draws 100 songs that carry any of the ticked genres, weighted by taste. When the queue runs out, “keep going” sticks to the same genres until you play something else. Untagged songs count under their artist's usual genres.

“For you” mixes

MIX	WHICH SONGS ARE IN THE RUNNING
Your mix	The whole library.
Fresh finds	Songs you've never played and haven't liked. The score still steers it towards what fits you (like Massive Attack for Sam).
Rediscover	Songs you liked or played at least twice, but not in the last 30 days.
Genre mixes (up to 6)	Your best-liked genres that have at least 8 songs; one mix each, songs of that genre only. A brand new listener gets the library's biggest genres.
Artist radios (up to 3)	Your top 3 artists. The pool is the artist's songs plus the songs of their 12 closest co-listened artists (and, if that's under twice the mix size, the artist's main genre). The artist's own songs get +0.3 so the mix stays recognisably theirs.

Each mix shows 30 songs on the page; pressing play makes a fresh 100-song version. Mixes with fewer than 5 songs are hidden. Every refresh is a new draw, so mixes change each visit.

Start radio (“More like this”)

Starts with the song you picked, followed by 50 songs like it. Here the main ingredient is how **similar** each song is to the seed, with your taste as a nudge:

```
similarity = 0.45 × artist   same artist 1, otherwise their co-listening similarity
           + 0.40 × genres   shared genres ÷ all genres of both
           + 0.15 × era      1 - decades apart ÷ 30 years (0.3 if unknown)
candidate  = similarity + 0.25 × your score
```

Songs with similarity 0.05 or less are left out. The best 400 candidates go into a much stricter draw (temperature 0.12 instead of 0.3) so the radio stays on topic, and the seed's own artist may fill at most a quarter of the list (12 of 50): the point is to find *other* artists like it.

Example, radio from Sam's “Paranoid Android” (A):

SONG	ARTIST	GENRE	ERA	SIMILARITY	CANDIDATE VALUE
B Radiohead “Idioteque”	1.00	1	0.67	0.95	1.10
C Massive Attack “Teardrop”	0.40	0	1.00	0.33	0.43
D Portishead “Glory Box”	0.20	0	1.00	0.24	0.21
__E__ Taylor Swift “Blank Space”	*0.00*	*0*	*0.33*	*0.05*	*left out (≤ 0.05)*
__F__ Miles Davis “So What”	*0.00*	*0*	*0.00*	*0.00*	*left out (≤ 0.05)*
G Muse “Hysteria”	0.30	1	0.67	0.64	0.69

“Idioteque” leads (same artist and genre). Muse's “Hysteria” comes next on genre and co-listening. Massive Attack makes it through co-listening and the shared decade. Miles Davis (no link, 40 years apart) and Taylor Swift (exactly 0.05) never appear in this radio.

10 All the numbers in one place

SETTING	VALUE	WHAT IT DOES
Play weight / half-life	+1.0 / 120 days	Each counted play, halving in value every 120 days
Like weight	+4.0	Each song in Liked Songs, never fades
Playlist weight	+0.75	Each song on your own playlists, once
Skip weight / half-life / window	−0.6 / 21 d / 90 d	Next pressed after 2 s and before the play counted
Skips to drop a song	2.5	Faded skip total that blocks a song (unless liked)
Dislike weight	−3.0	Taken from the disliked song's artists, genres and decade
Era share	½	A signal counts half towards the decade
Squeeze	tanh(1.5 × total ÷ typical)	typical = mean of your top 10 totals, at least 1; a song's own value is tanh(total ÷ 3)
Score mix	0.30 / 0.25 / 0.08 / 0.17 / 0.20	artist / genre / era / co-listening / the song itself
Just-played penalty	$0.8 \cdot e^{-h/6} + 0.25 \cdot e^{-h/72}$	h = hours since you last heard the song
Recent-skip penalty	0.25 × faded skips	
Shuffle temperature	0.3	weight = $e^{score/0.3}$; +0.9 ≈ 20× a neutral song
Radio temperature	0.12	Stricter, keeps “More like this” on topic
Artist spacing	3 songs	An artist can't return while it's among the last 3 songs
Share per artist / album	⅛ of the list (min 2)	Eased only when nothing else is left
Co-listening session gap / reach	30 min / 3 songs	Weights 1, ½, ⅓; top 40 neighbours per artist; last year of plays; rebuilt ≤ every 10 min
Similarity (radio)	0.45 / 0.40 / 0.15	artist / shared genres / era; minimum 0.05; own artist ≤ ¼
History read	400 days	Up to 20,000 newest plays per listener
Mixes	3 + 6 + 3	Your mix, Fresh finds, Rediscover + genre mixes (≥ 8 songs) + artist radios; hidden under 5 songs
Rediscover	30 days	Liked or ≥ 2 plays, not heard in the last month

Where it lives

- `server/crates/library/src/taste.rs` : the engine (catalog, co-listening, profile, score, draw, spreading, mixes, radio).
- `server/crates/api/src/routes/taste.rs` : the `/api/taste/…` endpoints (shuffle, mixes, mix, like-this, genres, profile, skip, dislikes) and what is read from the database.
- `web-rs/src/player/taste.rs` and `queue.rs` : the player side (dislike button, skip reporting, smart order, Start radio, Genre shuffle, keep going).